SYN-ACK WIRE

SEVEN SYNDICATES. ONE REGION. EVERY NODE FOR SALE.

USER MANUAL

FOR THE HEATHKIT H89

HEATHKIT H89 / ZENITH Z-89 // FUJINET

2-6 NETRUNNERS // BOTS FILL EMPTY SEATS

AN UNOFFICIAL CYBERPUNK RE-SKIN OF ACQUIRE BY SID SACKSON.

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01 // THE NETWORK GRID

THE SETTING // THE OBJECT // THE END

You are a **Netrunner**. You **jack-in** to the region's **grid**, examine your **exploit kit** and search for unpatched **nodes** to **spike**. When two spikes are linked, you've gained access. Syndicates want control of the grid and will pay a **finder's cut** in their **crypto** for this access. This becomes their **hive**. Hives spread across the grid with each new spike. As a hive grows, so does its crypto's value.

But syndicates do not share. When two hives collide, the bigger one **hivejacks** the smaller, absorbing it. The smaller hive's top two crypto backers make bank. Hope that you're one of them.

Eventually, the hives grow too big to remain unnoticed. The carriers catch on and **null-route** the entire region. Your gig is up. When the routes go dark, the netrunner holding the most ¥ wins.

You've learned the hard way: the real ¥ is not in brokering spikes, but in playing the syndicates against each other.

> GAME DESCRIPTION

SYN-ACK-WIRE is a strategic, turn-based, online multiplayer game for FujiNet-enabled computers set in a cyberpunk world. The table seats two to six players, including bots, so you can play solo. Take turns, spiking the network and dabbling in crypto, until an end-of-game condition exists.

> OBJECT OF THE GAME

Have the most new yen (¥) when the region is null-routed.

> END GAME CONDITIONS

Play can end when:

1. Any hive has become a Total Compromise by spreading to 41+ nodes.
2. All hives have become immune to hivejacks (**Black ICE**) by spreading to 11+ nodes. (automatic)

02 // JACK-IN

REGION // STATUS // HANDLE

SYN-ACK WIRE runs under Heath/Zenith CP/M 2.2.04 on an H89 with the H89 FujiNet card. There is no FujiNet lobby for the H89 yet: you start CP/M, then the game. The H89 always starts CP/M from its own drive; the FujiNet's drives join it once CP/M runs.

> FROM THE FUJINET

You need the *FujiNet for CP/M* diskette, with CONFIG and FNDISK, and a 62K CP/M system: the *FujiNet Operation Manual for the Heathkit H89* has both.

1. Start CP/M.
2. Run CONFIG (`B:CONFIG`). Pick a free host slot, push E and type the server: `syn-ack-wire.fly.dev`.
3. Open that host slot and push RETURN on the game's disk, `syn-ack-wire-h89.h8d`. Pick drive slot 1 and push RETURN again, for read only. Push ESC to leave CONFIG.
4. Run `B:FNDISK`: drive slot 1 becomes drive `D:`.
5. Type `D:` and press RETURN, then `SYNACK` and RETURN.

Change to `D:` first: the game reads its instructions from the drive CP/M is logged in to.

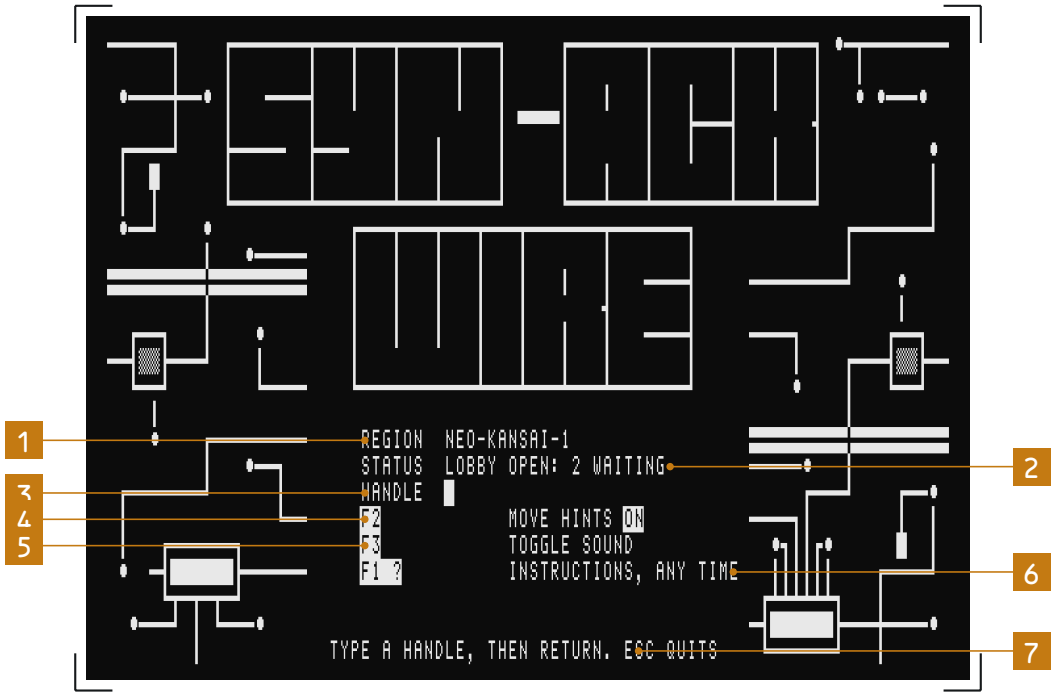
> FROM A DISKETTE

The same image, written to a hard-sectored diskette, starts CP/M by itself, and needs no FNDISK: put it in the first H17 drive, type B and RETURN at the monitor's `H:` prompt, then `SYNACK` and RETURN at `A>`. Or copy the game to a CP/M disk of your own, with `PIP A:=D:SYNACK.*`.

> THE TITLE SCREEN

The title screen dials the game's server, CHIBA-GRID-7, as soon as it appears. Type a handle, up to 8 characters, and press RETURN.

CAPTURE 01 TITLE SCREEN LOBBY OPEN



1 **Region.** The server's grid: one game at a time, up to six seats.

2 **Status.** Whether you can join, and how: LOBBY OPEN, IN PROGRESS, TABLE FULL. It stays up to date while you type.

3 **Handle.** Your name at the table.

4 F2 turns the move hints on or off.

5 F3 turns the sound on or off.

6 F1 or ? opens the instructions, here or in the middle of a game.

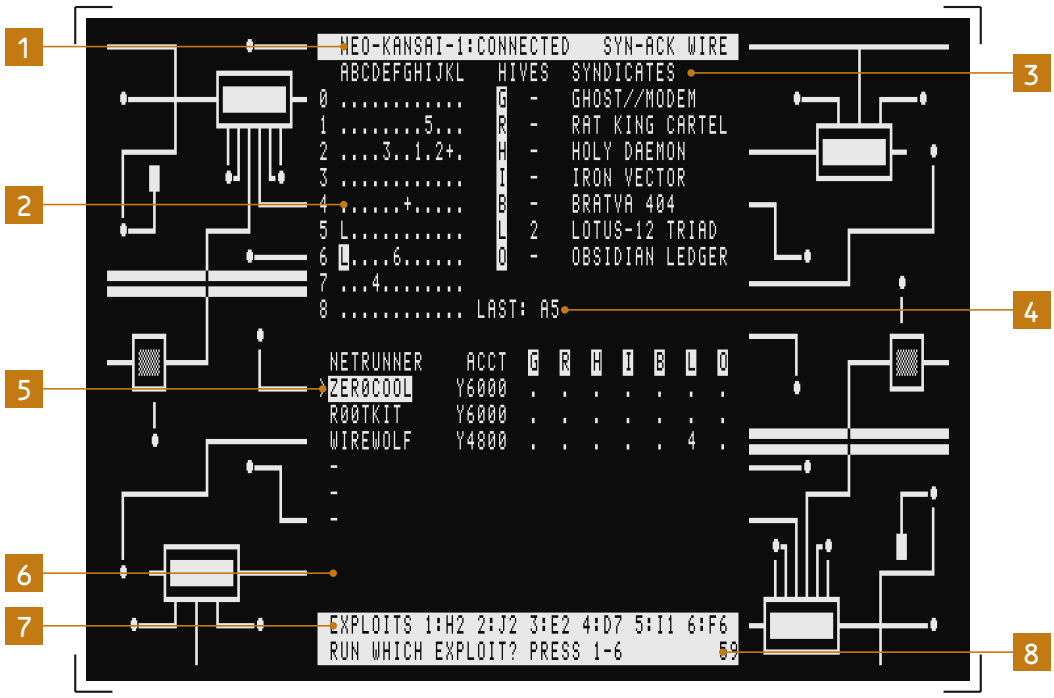
7 **The prompt line.** What the screen expects from you next; a refusal, such as a handle already taken, shows here in reverse.

03 // THE SCREEN

THE H19 // 80 x 24 // THE GAME IN 40

Everything you need is on one screen: the H19's middle 40 columns, with a circuit board drawn in its graphics characters at the sides. This is early in a game, and it's your turn: pick an exploit to run.

CAPTURE 02 SPIKE PROMPT 3 SEATS EARLY GAME



1 Title bar. The region and the link. It reads NO SIGNAL when the server goes quiet.

2 The grid. Reverse letters are hives; **+** is a lone spike. A digit marks a node an exploit in your exploit kit can run on. The last move flashes for a few seconds.

3 Hives. Each syndicate's letter, hive size and name. While you buy, its price and how many the Darkpool has left.

4 Last move. The node spiked most recently.

5 Netrunners. **>** marks you, ***** a bot, **?** someone away. Account, then crypto held. A reverse name has the turn.

6 The log. The last three things that happened at the table.

7 Your exploit kit. Your 6 exploits and the nodes they run on.

8 The prompt. What you're asked, with seconds left on the shot clock. The two bottom lines turn reverse when the decision is yours. Here: press 1–6.

04 // KEYS

THE H19 KEYBOARD // FUNCTION KEYS // NO MOUSE

SYN-ACK WIRE is played from the keyboard. Each question on the prompt line names the keys it takes; this chapter has them all in one place. The game checks an answer before sending it and says on the prompt line what's wrong, such as NOT ENOUGH NEW YEN.

> ANY TIME

F1 ?	Open the instructions, on the title screen or in the middle of a game.
F2	Move hints on or off: the digits on the grid that show where each exploit in your exploit kit can run.
F3	Sound on or off: the H19's bell. It rings to say it's on.

> THE TITLE SCREEN

A–Z 0–9	Type your handle, up to 8 characters.
BACK SPACE	Rub out the last character.
RETURN	Jack-in.
ESC	Quit to CP/M: Y to confirm, any other key to stay.

> THE INSTRUCTIONS

> →	Next page.
< ←	Previous page.
ESC	Back to where you were. RETURN, F1 and ? do the same.

> HOSTING

2–6	Start the game with that many seats. Bots take the seats nobody has joined.

> YOUR TURN

1–6	Spike: run that exploit from your exploit kit. A reverse one can't run now; press it and the prompt line says why.
A–Z	Infiltration: the letter of the syndicate to broker the new hive to.
A–Z	A tied hivejack: the letter of the hive that survives, or is absorbed next.
A–Z	Buy crypto: type the syndicate letters, up to 3, in any mix of the hives on the grid. One letter per crypto: L buys one, L L two, L L L three of the same, L G one each of two, L G 0 one each of three.
BACK SPACE	Take back the last crypto in a buy. DELETE does the same.
RETURN	Confirm the buy. RETURN alone buys nothing.
Y N	When the game may end: null-route the region now, or play on. RETURN plays on too.

> WHEN YOUR HIVE IS ABSORBED

0–9 RETURN	First, how many crypto to cash out; then how many to swap, an even number. The rest you HODL. BACK SPACE rubs out a digit.

> LEAVING

ESC Y	Log off for good. A bot takes over your seat at once, and you're back on the title screen, where ESC again quits to CP/M. Any key but Y stays.

05 // WHAT'S IN THE GAME

THE GRID // THE SYNDICATES // SETUP

SYN-ACK WIRE is played on a computer; a game server manages everything below.

The Grid 108 nodes in 9 rows (0–8) and 12 columns (A–L). A node is named column first, then row: A0 is the top-left corner, L8 the bottom-right.

Exploits 108 of them, one for each node. An exploit can only be run against the node it is written for. Each netrunner holds 6 exploits, called their **exploit kit**. Your exploit kit is secret.

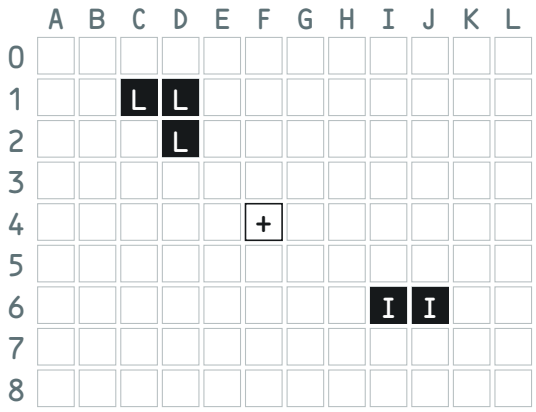
Syndicates Seven crime syndicates, each with its own crypto. Only one hive per syndicate can be on the grid at a time.

New yen (¥) The region's currency. Each netrunner starts with ¥6,000.

Crypto 25 of each syndicate's crypto. Crypto's value in new yen increases as hives spread. Crypto not owned by the netrunners is held by the Darkpool.

The Darkpool The bank. It sells crypto, buys it back, and pays out bonuses. There are no loans.

> READING THE GRID



Each syndicate is shown by its letter. **+**: a lone **spike** (F4). **L**: Lotus-12 Triad, a hive of 3 nodes (C1, D1, D2). **I**: Iron Vector, a hive of 2 nodes (I6, J6). Only up, down, left and right count as touching; diagonals never do.

> THE SEVEN SYNDICATES

Every syndicate's crypto follows the same rules. What differs is the **tier**: how much its crypto is worth for a hive of a given size. Low-tier crews are cheap to buy into; high-tier powers are expensive but pay big. The letter is how the syndicate appears on the grid.

	Syndicate	Tier	Word on the wire
G	Ghost//Modem	Low	Old-school phreakers squatting in dead telco switches. Able to whistle 2600 Hz.
R	Rat King Cartel	Low	Data smugglers in the maintenance tunnels. Sneaker Net.
H	Church of the Holy Daemon	Mid	Fanatical believers in the original init process. Prayers scheduled in crontab.
I	Iron Vector	Mid	Paramilitary arms brokers. Drone swarms for hire. No refunds.
B	Bratva 404	Mid	Eastern-bloc mob. Double-crossers become "not found."
L	Lotus-12 Triad	High	Hong Kong triad. Every favor a cell. Every cell a formula.
O	Obsidian Ledger	High	Faceless cabal of former central bankers. They run the Darkpool.

> SETUP

1. Every netrunner gets ¥6,000. The Darkpool holds all the crypto at the start.
2. **Turn order.** Each netrunner is dealt one exploit, and all are automatically spiked onto the grid at once. The netrunner whose spike is in the lowest-numbered row goes first; if two share a row, the one closer to column **A** wins the tie. Play then continues around the table in seat order. These opening spikes stay on the grid and do not form hives, even if they happen to touch.
3. Each netrunner is dealt 6 exploits into their exploit kit.

06 // PLAYING A TURN

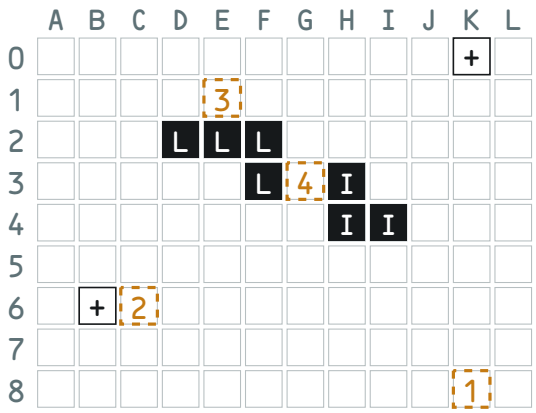
SPIKE // BUY // DRAW

On your turn, do these three things in order:

1. **Spike a node.** Run one exploit from your exploit kit and place a spike on its node.
2. **Buy crypto.** Use new yen (¥) in your account to buy up to 3 crypto from the Darkpool (optional). If your balance is too low, you won't see this option.
3. **Pick up a fresh exploit.** Draw a new exploit to bring your exploit kit back to 6. If you have no playable exploits, a fresh exploit kit will be provided.

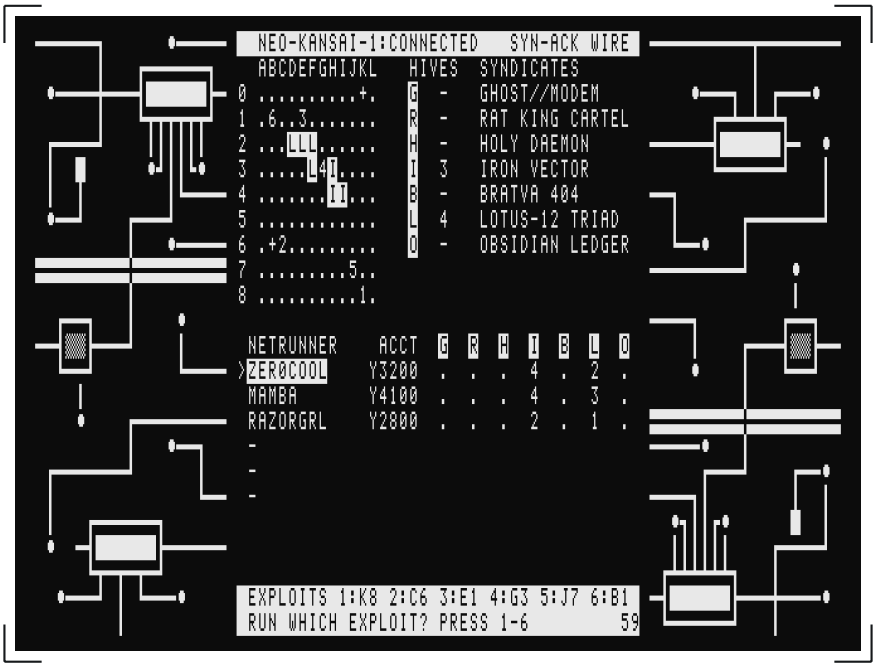
> 1. SPIKE A NODE

Where your spike lands decides what happens. It will do exactly one of these:



1. **Nothing.** `K8` touches nothing. The spike just sits there.
2. **Infiltration.** `C6` touches the lone spike on `B6`. A new hive is born.
3. **Spread.** `E1` touches the Lotus-12 Triad hive. It grows to 5 nodes.
4. **Hivejack.** `G3` touches both Lotus-12 Triad (4 nodes) and Iron Vector (3 nodes). Lotus-12 Triad absorbs Iron Vector.

CAPTURE SPIKE A NODE



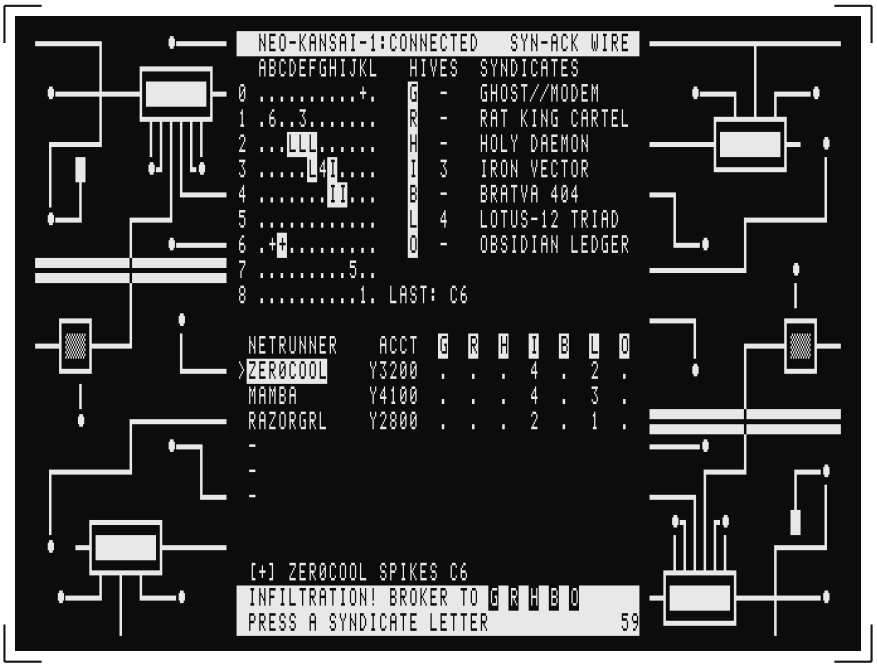
ON THE H89 The drawing above, as ZER0COOL sees it. The move hints 1–4 sit on its four targets: they are the exploits in your exploit kit that run there, as the info line lists them. Press the one to run.

> INFILTRATION: FOUNDING A HIVE

If your spike touches one or more lone spikes, and no hive, you have founded a new **hive**. Choose any syndicate that does not have a hive on the grid and **broker** the hive to it. Your spike and every lone spike connected to it become that syndicate's hive.

As your **finder's cut**, take 1 crypto of that syndicate from the Darkpool for free. If the Darkpool has none left, you get nothing.

CAPTURE INFILTRATION



ON THE H89 Target 2: ZER0COOL ran **C6**, next to the lone spike on **B6**. The info line lists, in reverse, the syndicates with no hive on the grid: press one of those letters to broker the new hive.

BLOCKED EXPLOITS

If all seven syndicates already have hives on the grid, an exploit that would found an eighth hive cannot be run. It stays in your exploit kit until a syndicate leaves the grid. The game marks these for you.

> SPREAD: GROWING A HIVE

If your spike touches exactly one hive, it joins that hive. Any lone spikes connected to your spike join too. The hive is now bigger, and its crypto is worth more.

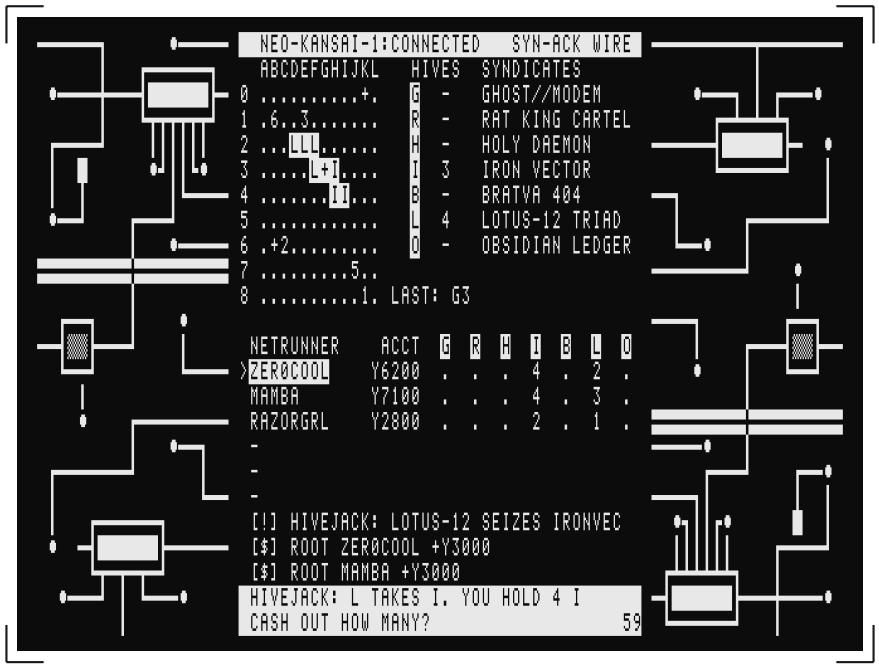
> HIVEJACK: MERGING HIVES

If your spike touches two or more different hives, they collide and the biggest one **hivejacks** the others. Your new spike does not count toward any hive's size while deciding this. Work through these steps:

1. **Who absorbs whom.** The hive with more nodes is the **absorber**; the smaller is **absorbed**. If they are the same size, you choose which syndicate survives.
2. **Find the ROOT and SUDO.** Look at who holds the absorbed syndicate's crypto. The netrunner with the most is **ROOT**; the one with the second most is **SUDO**.
3. **Payouts.** The Darkpool pays the ROOT and SUDO bonuses from the reference chart, using the absorbed hive's size just before the hivejack. See *Splitting payouts* below for ties.

4. **Crypto decisions.** Starting with you and going around the table, every netrunner holding the absorbed syndicate's crypto chooses what to do with it, in any mix:
   - **Cash out:** sell crypto to the Darkpool at the absorbed syndicate's price (its size before the hivejack).
   - **Swap:** trade 2 absorbed crypto for 1 crypto of the absorber, as long as the Darkpool has absorber crypto left.
   - **HODL:** keep the crypto. It is now **flatlined** and worthless, unless that syndicate is brokered a new hive later, when it comes **back from the flatline.**
5. **Clean up.** The absorbed hive's nodes, your spike, and any connected lone spikes all join the absorber. The absorbed syndicate leaves the grid and can be brokered again later.

CAPTURE HIVEJACK: A TIED ROOT



ON THE H89 The example below, as ZER0COOL sees it. The spike on **G3** joined Lotus-12 Triad (4 nodes) and Iron Vector (3); the log shows the tied ROOT paid. The info line says how many of the absorbed crypto you hold: type how many to cash out and press RETURN, then how many to swap, and RETURN again.

EXAMPLE: A TIED ROOT

Iron Vector (3 nodes, mid tier) is hivejacked. At 3 nodes, mid-tier crypto is ¥400, so the ROOT bonus is ¥4,000 and SUDO is ¥2,000. MAMBA and ZER0COOL each hold 4 Iron Vector crypto; RAZORGRL holds 2. MAMBA and ZER0COOL tie for ROOT, so they share ROOT + SUDO: ¥6,000 split two ways, ¥3,000 each.

Nobody is paid SUDO, so RAZORGRL gets nothing.
Then, in turn order, MAMBA chooses to cash out 2 of them for ¥800 and swap the other 2 for 1 Lotus-12 Triad crypto.

> SPLITTING PAYOUTS

- **Tie for ROOT:** add the ROOT and SUDO bonuses together and split them evenly among the tied netrunners, rounding each share *up* to the next ¥100. No SUDO bonus is paid.
- **Tie for SUDO:** split the SUDO bonus evenly among the tied netrunners, rounding each share up to the next ¥100.
- **One netrunner is the only holder:** they collect both ROOT and SUDO.
- **Nobody holds the crypto:** nothing is paid.

EXAMPLE: ROUNDING UP

A 2-node mid-tier hive is absorbed (ROOT ¥3,000, SUDO ¥1,500). Two netrunners tie for ROOT: ¥4,500 split two ways is ¥2,250, rounded up to ¥2,300 each.

> BLACK ICE: SAFE HIVES

Once a hive reaches **11 or more nodes**, its syndicate wraps it in **Black ICE**. A Black ICE hive can never be absorbed. It can still absorb smaller hives that are not Black ICE.

An exploit that would join two Black ICE hives can never be run. It is **patched out.** The game marks it for you.

> MULTI-WAY HIVEJACKS

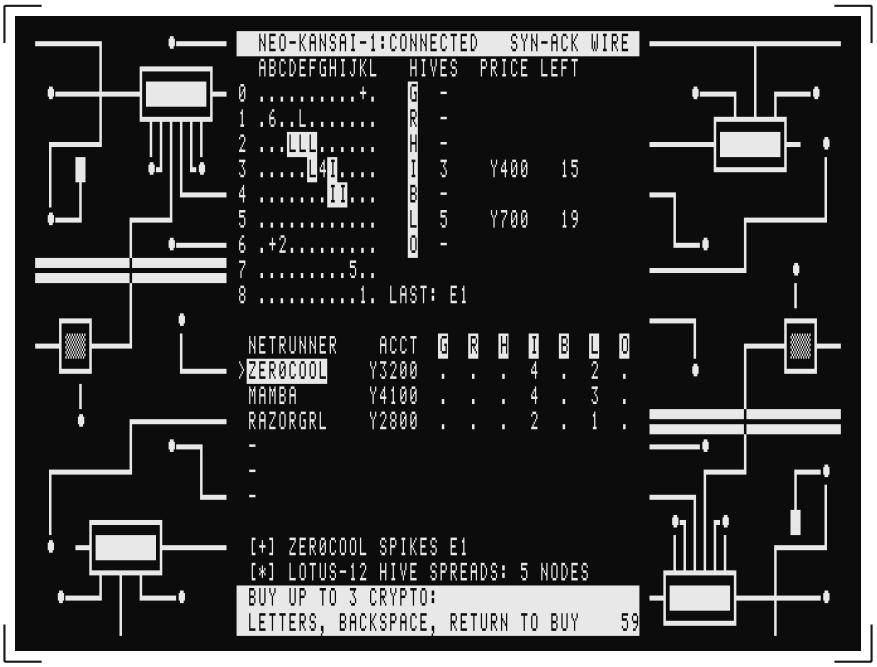
If your spike touches three or four hives at once, resolve them one at a time. The largest hive absorbs the second largest (do all five steps), then the now-bigger absorber takes on the next largest, and so on. If two of the smaller hives are tied in size, you choose which is absorbed first. If the two largest are tied, you choose the absorber.

> 2. BUY CRYPTO

After placing your spike, you may buy up to 3 crypto from the Darkpool, in any mix of syndicates. You can only buy crypto of syndicates that have a hive on the grid. The price is set by the hive's current size (see the reference chart), so a spread or hivejack you just caused changes the price before you buy.

You cannot buy more than the Darkpool has, and you cannot spend money you do not have. There are no loans.

CAPTURE BUY CRYPTO



ON THE H89 Target 3: `E1` spread Lotus-12 Triad to 5 nodes, and its price is already up. While you buy, the hives panel shows each price and how many the Darkpool has left. Type up to 3 letters, in any mix (`LLI` buys two Lotus-12 Triad and one Iron Vector), BACK SPACE to take one back, RETURN to buy.

> 3. PICK UP A FRESH EXPLOIT

Draw one exploit so your exploit kit is back to 6. If no exploits are left to draw, skip this step.

BURN THE KIT

If, at the start of your turn, none of the exploits in your exploit kit can be run, reveal your exploit kit, discard it, and draw 6 new exploits. Then take your turn as normal. The game does this automatically.

07 // ENDING THE GAME

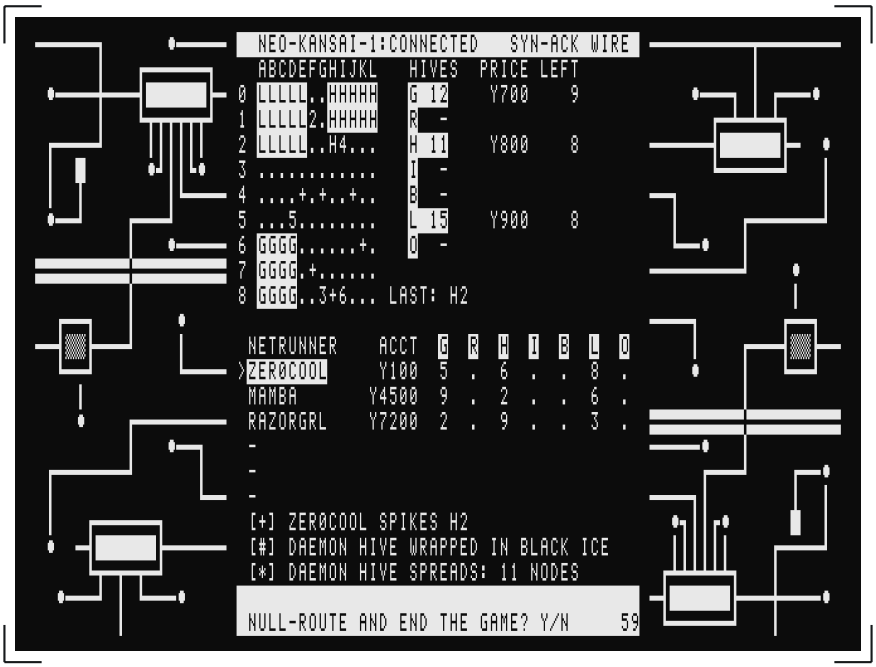
TOTAL COMPROMISE // LOCKDOWN // CASH OUT

After placing your spike, if either of these is true, you may **call the null-route:**

- **Total Compromise:** any hive has 41 or more nodes.
- **Lockdown:** every hive on the grid is Black ICE.

Calling it is your choice; you may keep playing instead. If you call it, finish your turn (you may still buy crypto). Then the carriers null-route the region and the game ends.

CAPTURE LOCKDOWN: THE QUESTION



ON THE H89 ZER0COOL's spike on H2 spread Church of the Holy Daemon to 11 nodes: Black ICE, and with Lotus-12 Triad and Ghost//Modem already Black ICE, every hive is. With too little ¥ for any crypto, the prompt is the question alone: Y calls the null-route, N or RETURN plays on.

```
[!]  NULL-ROUTE
NEO-KANSAI-1:  UNJACK!
```

GRID STALL

If a full round goes by in which no

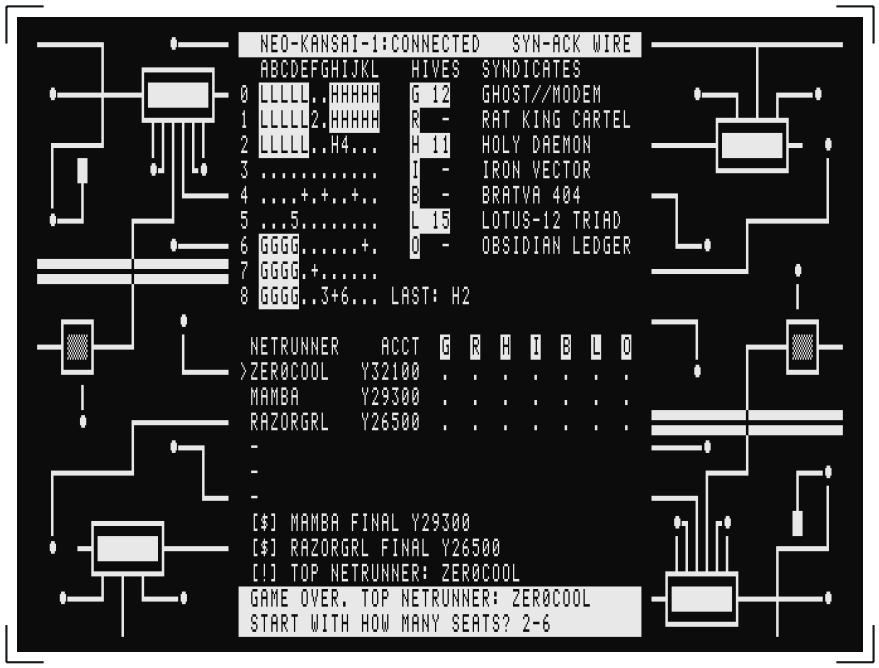
netrunner can run an exploit (every exploit kit is empty or holds only blocked or patched out exploits), the grid can never change again. The carriers seal the region and the game ends at once, without anyone calling it. Go straight to the final cash out.

> THE FINAL CASH OUT

1. Pay ROOT and SUDO bonuses for every hive still on the grid, exactly as in a hivejack.
2. Every netrunner sells all their crypto to the Darkpool at its current price, starting with the smallest hive.
3. Flatlined crypto (from syndicates not on the grid) is worth nothing.

The netrunner with the most ¥ wins. Ties share the win.

CAPTURE THE END OF A GAME



ON THE H89 ZER0COOL called it. ROOT and SUDO were paid for every hive, all crypto was sold, and the log ends with each netrunner's final ¥ and the top netrunner. The table is a lobby again: the host's prompt asks for the seats of a rematch.

08 // HOSTING, JOINING AND LEAVING

THE LOBBY // THE HOST // SEATS AND BOTS

A SYN-ACK WIRE server holds one game at a time: one region, up to 6 seats. Every seat with no netrunner at the keyboard is played by a bot, so a game never waits long on someone who isn't there. Times given here are the usual ones; whoever runs the server can change them.

> THE TITLE SCREEN'S STATUS

The title screen dials the server as soon as it appears, and its status line stays up to date while you type your handle.

`LOBBY OPEN: 2 WAITING` No game yet; 2 netrunners wait to play. RETURN takes a seat in the lobby.

`IN PROGRESS: TAKE OVER A BOT` A game is on, and a bot holds at least one seat. RETURN takes over a bot.

`FULL: DROPPED PLAYERS REJOIN` A game is on, every seat is a netrunner's, and some have dropped. Only they get back in.

`TABLE FULL` 6 netrunners are playing. RETURN does nothing.

`NO CARRIER` The server didn't answer. The game dials again.

People at the title screen don't hold a seat yet, and nobody waits for them.

> HANDLES

Your handle is your name at the table, up to 8 letters and digits. Only one seat can use it: a handle already at the table, a bot's included, is refused with `THAT HANDLE IS TAKEN`, and you can type another. Upper and lower case count as the same. The one exception is coming back to your own seat (see *Rejoining*).

> HOSTING

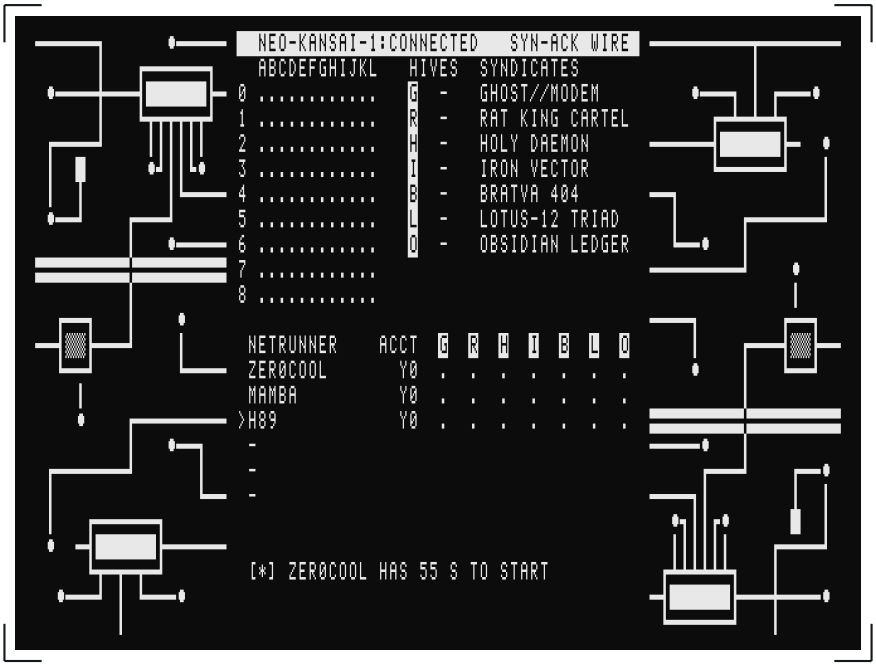
- **Who hosts:** the first netrunner into an empty lobby. Your prompt line reads `START WITH HOW MANY SEATS? 2-6`.
- **Starting:** press 2–6. The game starts at once; seats nobody has joined go to bots, named like NULLPTR and WIREWOLF. You can't choose fewer seats than there are netrunners waiting: with four in the lobby, the server says `[-] START WITH 4-6 SEATS`.
- **Who might come:** the seat rows show who has joined, and the log counts the people at the title screen, who may be about to: `[*] 2 AT THE DOOR`. It changes as they arrive, join or leave.

- **No hurry alone:** with nobody else seated, take as long as you like.
- **The host clock:** once someone else is waiting, you have 60 seconds to either start the game or let the host duty pass to the next netrunner. Letting it pass is fine: you keep your seat and play when the new host starts. The seconds count down at the right of your prompt line, and the others read `[*] ZEROCOOL HAS 60 S TO START`. When the clock runs out, everyone reads `[*] MAMBA IS THE HOST NOW`, and the new host has a fresh 60 seconds.
- **An absent host,** on autopilot, hands over at once when an active netrunner joins. A host who leaves hands over to the next netrunner in the lobby.

> JOINING A LOBBY

Type your handle and press RETURN. You take the next empty seat and appear in everyone's seat rows. If the host clock is running, the log tells you how long the host has left. You play when the host starts.

CAPTURE JOINING A LOBBY



ON THE H89 ZER0COOL hosts and MAMBA is waiting; the H89 has just joined. The host clock started when MAMBA arrived, and every newcomer is told how long the host has left.

> JACK-IN: JOINING A GAME IN PROGRESS

A newcomer to a game in progress takes over the bot seat worth least (cash plus crypto), as it stands: its ¥, crypto and exploit kit are yours now, and your handle replaces the bot's. You read `[*] GAME IN PROGRESS: TAKING OVER BOT`, and everyone reads `[+] ZEROCOOL JACKS IN FOR HONEYPOT`. A decision the bot is

already making stays the bot's; you play from the next one.

- Two people going for the last bot seat: the first RETURN wins. The other reads TOO LATE: THE LAST SEAT WAS TAKEN and stays on the title screen.
- The server may disable the jack in function for a game among friends. A game in progress then shows TABLE FULL.

> THE SHOT CLOCK

- **The shot clock:** each decision has 60 seconds, counted down at the right of the prompt line. At 0 your seat's bot makes that one decision. The seat stays yours.
- **Autopilot:** after two decisions in a row made by the bot, you are on autopilot. Everyone reads [?] MAMBA ON AUTOPILOT, and your seat shows ?. You still get every prompt, but on the bot's clock, about 10 seconds, so the game doesn't drag.
- **Back from autopilot:** answer one prompt in time. The full clock comes back, and everyone reads [*] MAMBA JACKS BACK IN.

> LOG OFF: LEAVING ON PURPOSE

Any time after joining, ESC asks LOG OFF NOW? Y/N on the prompt line.

- Y leaves for good. In a game, a bot takes your seat at once, under a bot's name, and plays at bot speed: nobody waits on your shot clock. Everyone reads [-] GL1TCH TAKES OVER FROM MAMBA. Your old seat is now an ordinary bot seat, which anyone, you too, can jack in for. In the lobby, your seat is simply freed.
- Any other key stays, and the question you were answering comes back.

You land on the title screen with your handle filled in.

> SWITCHING OFF, OR LOSING THE CONNECTION

- **In the lobby:** your seat is freed at once. If you hosted, the next netrunner does.
- **In a game:** your seat is kept for you and shown ?; its bot plays every decision until you are back. Your ¥, crypto and exploit kit stay with it.

A server notices a clean disconnect at once. A computer switched off, or a network that just stops, takes it up to about a minute; meanwhile the shot clock covers any decision.

> REJOINING

- **The connection drops:** after 20 seconds with nothing from the server, the title bar reads NO SIGNAL. At 30 seconds the game hangs up and dials again by itself. Once back, the server sends the whole game again, any decision waiting included. No key needed.

- **After a reboot:** type the same handle on the title screen and press RETURN. You get your own seat back, not a bot's, as long as it is shown away. If the server hasn't noticed the drop yet, your handle is still in use (THAT HANDLE IS TAKEN): wait a minute and try again.

> EVERYBODY GONE

- **Nobody connected:** the game pauses; bots don't play on for nobody. The first one back finds it as it was.
- **Paused too long:** after 10 minutes with nobody connected, the game is dropped, with no result, and the region is an empty lobby again.
- **Everyone on autopilot** for 10 minutes: the game is dropped too. Those still connected land in a fresh lobby, the first of them hosting, and read [!] GAME DROPPED: NOBODY PLAYING.

> AFTER THE NULL-ROUTE

When the game ends, everyone sees the final ¥ and the top netrunner. The table is a lobby again, with the same host (or the next netrunner still connected), whose prompt asks for the seats of a rematch. Netrunners still connected keep their seats; a newcomer may take any other chair. A netrunner who rejoins after the end sees the result.

09 // WHERE THE MONEY COMES FROM

FINDER'S CUT // ROOT AND SUDO // CASH

New netrunners often feel busy without knowing what they are building toward. It helps to know where the ¥ actually comes from.

IN ONE LINE

Exploits don't make money; crypto does. You get paid when a hive you lead is absorbed, or when the game ends. Running exploits is how you decide which hives get absorbed, and when.

> FOUNDING IS ABOUT POSITION, NOT MONEY

An infiltration earns only the finder's cut: one free crypto, worth ¥200–¥400 while the hive is small. Its real value is that you choose which syndicate comes up, and you are first to buy its crypto, at the lowest price it will ever have. Founding is the cheapest way to become ROOT of a hive.

> THE BIG MONEY: ROOT AND SUDO

- **The absorbed hive pays, not the absorber.** This is the part new netrunners get backwards. At a hivejack, only the syndicate being swallowed pays its ROOT and SUDO bonuses.
- **The bonuses are large.** ROOT pays ten times the crypto's price and SUDO five times. At a price of ¥600, ROOT is ¥6,000, as much as your whole starting stake.
- **So the classic play** is to lead a small or medium hive that sits next to a bigger one, then run the exploit that joins them, on your terms.

> THE REST: THE CRYPTO ITSELF

- **At a hivejack**, the absorbed crypto can be cashed out, swapped 2 for 1 into the absorber, or kept. A swap is how you climb into a big hive without spending ¥.
- **At the null-route**, every hive still on the grid pays ROOT and SUDO, and all crypto is sold at its final price. A Black ICE hive can never be absorbed, so it pays only now, but by now it is usually big.

> WHY IT IS HARD TO SEE: CASH IS THE REAL LIMIT

You start with ¥6,000 and may buy only 3 crypto a turn, and nothing you hold pays until a hivejack or the end. An early bonus counts twice: it pays you, and it pays for the crypto that wins the next fight for ROOT.

EXAMPLE: FINDER'S CUT VS. ROOT

Early on you found a low-tier hive: your finder's cut is worth ¥200. You keep buying, and when the hive has grown to 5 nodes you hold 4 of its crypto and nobody else holds more than 2. A larger hive sits one node away, and you hold that exploit. You run it: ¥5,000 for ROOT, then you cash out your 4 crypto at ¥500 each for ¥2,000 more. The infiltration was the doorway; the hivejack was the payday.

10 // TIPS

EARLY // MIDDLE // LATE

None of this is a rule. It is what experienced netrunners tend to know.

> EARLY: GET HIVES ON THE GRID

- **Found hives whenever you can.** The finder's cut is a free crypto, and you can buy 3 more on the same turn, while the price is at its lowest. That is a head start on ROOT that nobody else gets.
- **Small hives pay too.** A hive that gets hivejacked early still pays its ROOT and SUDO. Don't ignore a hive just because it will never be big.
- **Don't stretch yourself thin.** With only 3 crypto a turn, being ROOT of one syndicate is usually worth more than being third in three.

EXAMPLE: A QUICK FLIP

You found a low-tier hive of 2 nodes, take your finder's cut, and buy 3 more crypto at ¥200 each (¥600). Nobody else buys in. Next turn it is hivejacked while still at 2 nodes. As the only holder you collect ROOT and SUDO (¥3,000), then cash out your 4 crypto for ¥800. You spent ¥600 and got ¥3,800 back.

> MIDDLE: PICK YOUR MOMENT

- **Look at the holdings.** Crypto holdings are public. Before you broker a hive to a syndicate, or set off a hivejack, check who would be paid. Don't hand a rival a bonus.
- **One crypto more than the leader takes ROOT.** A tie for ROOT splits the money. If you can see you are tied, a single extra crypto takes the whole ROOT bonus.
- **Grow it before it's absorbed.** Payouts and cash out prices use the absorbed hive's size before the hivejack. If you hold its crypto, push it up to the next step of the price chart first. Check the chart: a low-tier hive pays the same at 6 nodes as at 10, so growing it from 6 to 10 earns you nothing.
- **Don't grow a rival's hive.** Every spread raises the price and moves a hive toward a hivejack or Black ICE. Make sure it is your hive you are helping.
- **Keep options in your exploit kit.** Some exploits touch nothing and just pass the turn quietly. Others found, spread, or hivejack. Holding a mix lets you choose when things happen instead of being forced.
- **Count exploits.** There is exactly one exploit for each node. If a key node stays empty turn after turn, someone may be holding its exploit for the right moment.

> HIVEJACK CHOICES

- **Swap when the absorbed crypto was cheap.** A swap is always 2 for 1, whatever the sizes. Two crypto worth ¥300 each (¥600 if you cash out) for one worth ¥800 is a good trade, and it might make you ROOT of the absorber.
- **HODL if you can bring it back.** If your exploit kit can found a new hive, keeping flatlined crypto and then brokering that syndicate again gives you a head start on ROOT. Selling now and buying back later costs turns.
- **Cash early, swap late.** In the first few hivejacks, new yen buys you more crypto. Later, one hive usually gets big enough to last the game. Get into it with a swap before the end, or make sure you hold ROOT somewhere else that is big.

> LATE: THE NULL-ROUTE

- **Black ICE hives survive.** At 11 nodes a hive can never be absorbed, so its ROOT bonus is settled at the final cash out. A hive one node short of Black ICE is the best target on the grid, for you or for a rival.
- **Flatlined crypto is worth nothing at the end.** HODLing late in the game is a gamble that the syndicate comes back in time.
- **Only call the null-route if you are ahead.** Add up your ¥, your crypto at current prices, and the bonuses

you would collect. If someone else comes out on top, keep playing.

- **Use the other netrunners' turns.** The shot clock only runs on your decisions, so plan your next spike while others move. With move hints on, each empty node your exploit kit can hit shows the number of the exploit that hits it.

Some of these tips are adapted from Robert Crowter-Jones's *Acquire* strategy notes (Elusive Meeple, 2018).

11 // FREQUENTLY ASKED QUESTIONS

THE RULES // THE H89

What if I run out of new yen? Keep running exploits. You just can't buy crypto until a payout or cash out puts ¥ back in your account.

Can I see other netrunners' crypto? Yes. Crypto holdings are public. Exploit kits are secret.

Does my free finder's cut count toward ROOT and SUDO? Yes. It counts like any other crypto you hold.

Can a flatlined syndicate come back? Yes. Once a syndicate leaves the grid it can be brokered a new hive. Any crypto you HODLed is worth its price again.

Do I have to call the null-route? No. If you are behind, keep playing and hope to catch up.

> ON THE H89

Which machines? A Heathkit H89, or a Zenith Z-89, with 64K, an H17 drive and the H89 FujiNet card, under Heath/Zenith CP/M 2.2.04. The game draws on the H19 terminal built into the H89: all its text and graphics characters.

Why does the screen take a moment to change? The H19 is a terminal on a serial line inside the H89, at 9600 bits a second. The game sends it only what changed, but a whole new screen, such as a page of the instructions, takes about a second to draw.

Can I play alone? Yes. Host a game and press 2: one bot plays you. Any seat nobody takes goes to a bot.

What are the digits on the grid? Move hints: each marks a node that an exploit in your exploit kit can run on, and the digit is its key. F2 turns them on or off.

What do * and ? after a name mean? * is a bot. ? is a netrunner who is away, or on autopilot; a bot plays for them meanwhile.

Can I turn the sound off? Yes: F3, any time. The bell rings to say it's back on.

What do I hear? The H19's bell: when it's your turn, at a hivejack, and at the null-route.

What is the art at the sides? Decoration: a circuit board, the same all game. It shows nothing the middle 40 columns don't.

Can I play against an Atari or a PC? Yes. The servers are the same for every machine: H89, Atari, Apple II, PC and CoCo players meet at the same tables.

Why was my handle refused? Someone at the table has it, a bot included. If it's your own seat after a reboot, the server may not have noticed the drop yet: try again in a minute.

What if everybody leaves? The game waits for 10 minutes with nobody connected, then it is dropped.

12 // GLOSSARY

EVERY TERM, FROM THE THEME

Netrunner You, the player.

Black ICE A hive of 11+ nodes. Can't be absorbed.

Broker Assign a newly founded hive to a syndicate.

Burn the kit Discard an unusable exploit kit and draw 6 new exploits.

Cash out Sell crypto to the Darkpool.

Darkpool The bank.

Exploit One card in your exploit kit, good for one node.

Exploit kit Your hand of 6 exploits.

Finder's cut The free crypto for founding a hive.

Flatlined Crypto of a syndicate not on the grid.

Grid The board: one region's network.

Hive A syndicate's connected group of nodes.

Hivejack One hive absorbing another.

HODL Keep absorbed crypto instead of selling or swapping.

Infiltration Founding a new hive.

Lockdown Every hive is Black ICE; the game may end.

Grid Stall A full round with no exploit run; the game ends.

Node One square of the grid.

Null-route The end of the game.

Patched out An exploit that can never be run.

ROOT Biggest holder of a syndicate's crypto; gets the big bonus.

Spike A placed exploit; a lone spike belongs to no hive.

Spread A hive growing by one or more nodes.

SUDO Second-biggest holder; gets the small bonus.

Swap Trade 2 absorbed crypto for 1 absorber crypto.

Total Compromise A hive of 41+ nodes; the game may end.

REF // CRYPTO PRICES AND PAYOUTS

FIND THE HIVE'S SIZE IN ITS TIER, THEN READ ACROSS

LOW (G R)	MID (H I B)	HIGH (L O)	PRICE	ROOT	SUDO
2	-	-	¥200	¥2,000	¥1,000
3	2	-	¥300	¥3,000	¥1,500
4	3	2	¥400	¥4,000	¥2,000
5	4	3	¥500	¥5,000	¥2,500
6-10	5	4	¥600	¥6,000	¥3,000
11-20	6-10	5	¥700	¥7,000	¥3,500
21-30	11-20	6-10	¥800	¥8,000	¥4,000
31-40	21-30	11-20	¥900	¥9,000	¥4,500
41+	31-40	21-30	¥1,000	¥10,000	¥5,000
-	41+	31-40	¥1,100	¥11,000	¥5,500
-	-	41+	¥1,200	¥12,000	¥6,000

RED FIRST BLACK ICE SIZE: THE HIVE CAN'T BE ABSORBED
ROOT 10 x CRYPTO PRICE
SUDO 5 x CRYPTO PRICE
41+ TOTAL COMPROMISE: THE NULL-ROUTE MAY BE CALLED

THE LETTERS: EACH SYNDICATE ON THE GRID, AND ITS TIER

G	Ghost//Modem	LOW	B	Bratva 404	MID
R	Rat King Cartel	LOW	L	Lotus-12 Triad	HIGH
H	Church of the Holy Daemon	MID	O	Obsidian Ledger	HIGH
I	Iron Vector	MID			